

Figure 1: How Chef works with AWS nodes



Figure 2: The Chef client

some service provider as well. It is used to deploy cookbooks to the nodes. There is an option of signing up for Opscode's Hosted Chef. Each node is registered with the Chef server, which distributes cookbooks to the nodes based on different configuration settings.

The Chef client is installed on the nodes; it contacts the Chef server, fetches cookbooks and executes them on nodes. The Chef client does the actual configuration. It polls the Chef server at regular intervals for the most recent recipes, and checks node compliance with the policy defined by the most recent recipes. If the node is out of sync, the Chef client runs the policy on the node to bring it up-to-date. A node can be a physical server, a virtual server or a virtual machine in a private cloud or public cloud environment. Many plug-ins are available for Knife that provide support for cloud-based environments. Amazon EC2, Amazon Virtual Private Cloud, VMware, Google Compute Engine, OpenStack, Rackspace and Microsoft Azure are supported cloud platforms or cloud-based environments. A Chef client is an agent that runs locally on every node hosted in different environments, and follows the steps shown in Figure 2.

## Chef, VMware, AWS and continuous delivery

### A Chef and VMware-based private cloud

Chef is the configuration management tool and an automation

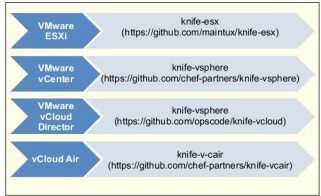


Figure 3: Chef plug-in for VMware

platform for installing, configuring and managing VMware infrastructure. It can be used to automate the entire stack including provisioning and de-provisioning VMs, templates and networks.

vCloud Air is a public cloud platform built on the foundation of vSphere. It is compatible with the on-premises data centre. It allows organisations to extend workloads into the cloud or to migrate existing onsite virtual machines to the public cloud. Chef can be used to automate migrations to vCloud Air. Chef has the capability of provisioning new vCloud Air VMs, vApps, templates or images, networks, NAT rules, firewall rules and public IP addresses.

VMware vCenter Server is a centralised management tool to manage virtualisation environments based on VMware vSphere. It can be used to manage multiple ESXi servers and virtual machines with a single console application. It provides enterprise level features such as high availability and fault tolerance with VMotion, Storage VMotion and Distributed Resource Scheduler. The *knife-esx* plug-in gives you full control of ESX-hosted virtual machines. Chef's *knife-vsphere* plug-in is used to provision and manage VMs with VMware vCenter. The *knife-vsphere* plug-in can be used to list data stores, resource pools and clusters; to list, clone, and delete VMs; perform specific actions by executing commands on deployed virtual machines; to customise or reconfigure vCPUs, VRAM, and IP addresses to clone a VM; and to clone and bootstrap Linux VMs and Windows VMs.

VMware ESXi is an enterprise-level operating system-independent hypervisor offered by VMware. It is a Type 1 hypervisor for guest virtual machines. It runs on a physical host without an underlying operating system support and hence it gives better performance than Type 2 hypervisors. Chef can be used to provision/de-provision or manage virtual machines hosted by ESXi. The *knife-esx* plug-in is used to integrate Chef with ESXi.

### Advantages of Chef in a VMware environment

There are many advantages of using Chef in a VMware environment as listed below: